STATEMENT OF PROBLEM

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When should you write a problem statement?

There are various situations in which you might have to write a problem statement.

In the business world, writing a problem statement is often the first step in kicking off an improvement project. In this case, the problem statement is usually a stand-alone document.

In academic research, writing a problem statement can help you understand the significance of your research problem.

It is often several paragraphs long and serves as the basis for your research proposal. Alternatively, it can be condensed into just a few sentences in your introduction.

Basic Writing Strategies for the Problem Statement

- The problem statement highlights the pressing issue the research intends to address.
- It should be concise and to the point. Researchers can follow a two-step approach:
- first, think about the content of the problem statement,
- Second, organize the writing framework.



How to Write an Effective Problem Statement | Elsevier

1

Before writing, clarify the following points:

- What is the reader's level of understanding of the research topic?

- How can the significance of the research be effectively conveyed to the reader?
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2

After addressing these two questions, you can organize the content according to the following structure:

- **Clarify what you aim to achieve with your research.**
- **Explore why the problem exists and explain how solving it helps reach the goal.**
- **Outline the potential impact of the research, such as possible outcomes, challenges, and benefits.**
- **Recommend a plan for your experiment that follows the rules of science.**
- **Explain the potential consequences if the problem is not resolved (if applicable).**

Three Important Parts of the Problem Statement

The content and length of the problem statement can vary depending on the type of research. Although there's no fixed format, it's helpful to include these three key parts:

- **Research Background**
- **Research Significance**
- **Solution**

•Research Background:

Explain clearly what problem your research focuses on. Describe how things would be better if this problem didn’t exist. Also, talk about what other researchers have tried to do about this problem and what still needs to be figured out.

• Research Significance:

Clarify the impact of the problem on the research field and society and analyze the cause of the problem. Explain who will benefit from solving the problem, thus demonstrating the relevance of the research and its contribution to the existing research system. To illustrate the relevance, consider aspects such as the geographical location or process where the problem occurs, the period during which it exists, and the severity of the problem.

•Solution:

Describe the research objective and the expected solution or results.



Research Topic:

The benefits of vitamin D supplementation on the immune system.

Problem Statement:

1.Review existing research on the role of vitamin D in the immune system, emphasizing the potential impacts of vitamin D deficiency on the human body.

2.List the obstacles encountered when trying to increase vitamin D levels in the body through supplements, and briefly mention the physiological or molecular mechanisms behind these obstacles.

3.Clarify feasible ways to overcome obstacles, such as new methods to promote the absorption of vitamin D in the intestine. Then, focus on the benefits of these methods, such as helping postmenopausal women with breast cancer improve their blood vitamin D levels.

Points to Note:

When crafting your problem statement, focus on essential details and avoid unnecessary information. Additionally, absolute terms such as "must" should be avoided.

Remember

Research in general is either:

- Observational (You are passive, taking measurements for the population)
- or Clinical trial where you deliver the intervention and examine its effects in a clinical trial.

How to Phrase Your PICO

Diagnosis	Are (is) _____ (I) more accurate in diagnosing _____ (P) compared with _____ (C) for _____ (O)?
Therapy	In _____(P), what is the effect of _____(I) on _____(O) compared with _____(C)?
Prognosis	Does _____ (I) influence _____ (O) in patients who have _____ (P)?
Etiology/Harm	Are ____ (P) who have _____ (I) at ___ (Increased/decreased) risk for/ of_____ (O) compared with _____ (P) with/without _____ (C)?
Prevention	For _____ (P) does the use of _____ (I) reduce the future risk of _____ (O) compared with _____ (C)?

TYPES OF RESEARCH QUESTION

-Prevalence/incidence

-Etiology

-Diagnosis

-Therapy/Prevention

-Prognosis

PREVALENCE

Prevalence (frequency) of a disease:

The proportion of people with the disease measured at one point in time. (cross-sectional)

importance of prevalence question???-

Incidence (rate): the proportion of people who are new cases of the disease within a specified period. (longitudinal)

PO

ETIOLOGY

The reverse of intervention questions because they deal with the harmful outcomes of an activity or exposure. (P: healthy, O: harm)

"For healthy individuals…
…having bruxism…
…..compared to those with no bruxism…
…… what is the risk of developing TMD?" Causation

But, if we survey how many TMD patients have bruxism, this is an association.

Association Doesn't mean Causation

PECO
PEO

DIAGNOSIS

"In patients with undiagnosed oral lesions...
...can clinical examination,...
... when compared to an oral biopsy,...
...effectively detect oral cancer?"

PIRT

Diagnostic studies I:index, R:reference test(control), T:target condition(outcome)

THERAPY(INTERVENTION)/ PREVENTION

“For completely edentulous patients…
…will antibiotic administration …
… When compared to no intervention,…
……increase implant stability for overdenture?” Intervention

“ For patients with high caries index
…..will fissure sealant.
…….as compared to varnish.
……..prevent the development of dental caries?” Prevention

PICO, PIO

PROGNOSIS

How likely is an outcome for a population with certain characteristics (risk factors)
“In patients with implants…
… who smokes…
… compared to patients who don’t smoke…

What is the proportion of implant loss at 10 years?”
In complete denture patients
…with a flat ridge
…..compared to those with normal ridge
……..what is the expected longevity of the denture

PICO, PIO




Thank
you